SEVEN ZONES. ONE CABLE PARTNER.

From the 230 kV switchyard to the portable power pad, each part of the campus asks something different of its cable. Here are the seven zones of the SK-3X1 campus and the five products that carry each one.

01 GRID CONNECTION

1 HV XLPE transmission cable, 230 kV
2 Overhead conductor (ACSR / ACSS)
3 Bare copper grounding conductor
4 Type TC-ER control cable, shielded
5 MV-105 shielded power cable

02 POWER ISLAND

1 ARMOR-X® MC-HL 15 kV
2 ARMOR-X® MC-HL 600 V
3 MV-105 15 kV, EPR 133%
4 VFD cable, symmetrical grounds, shielded
5 Instrumentation cable (PLTC / ITC)

03 BALANCE OF PLANT

1 MV-105 5 kV and 15 kV
2 Type TC-ER 600 V power
3 SIMpull® THHN / XHHW-2 building wire
4 Type TC-ER control, multiconductor
5 Fire alarm and life-safety cable

04 CARBON CAPTURE

1 MV-105 15 kV
2 ARMOR-X® MC-HL 15 kV
3 ARMOR-X® MC-HL 600 V
4 VFD cable
5 Instrumentation cable (PLTC / ITC)

05 FUEL

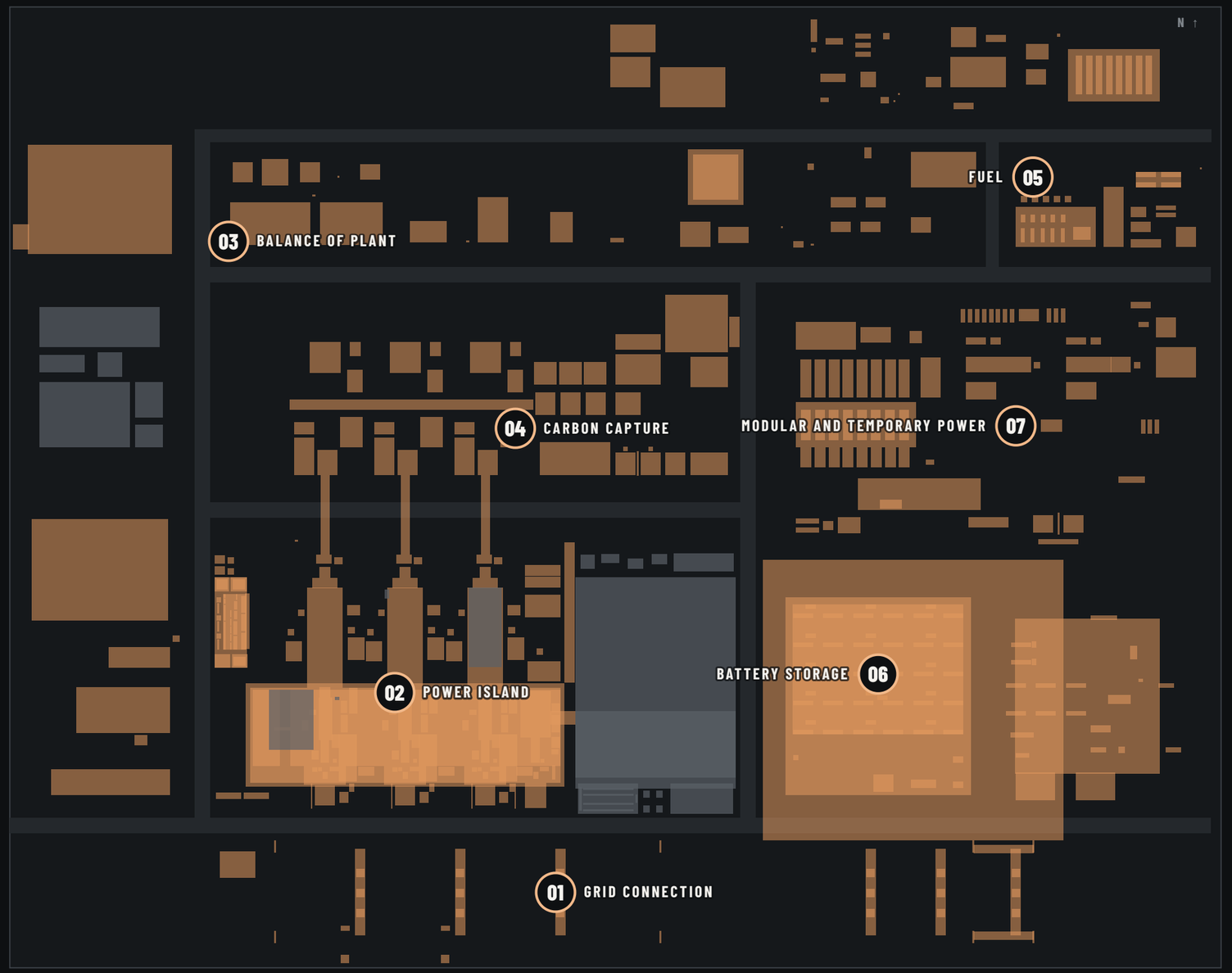
1 ARMOR-X® MC-HL 600 V
2 Type TC-ER-HL power and control
3 Armored instrumentation cable
4 MV-105 5-15 kV
5 Bare copper grounding and bonding

06 BATTERY STORAGE

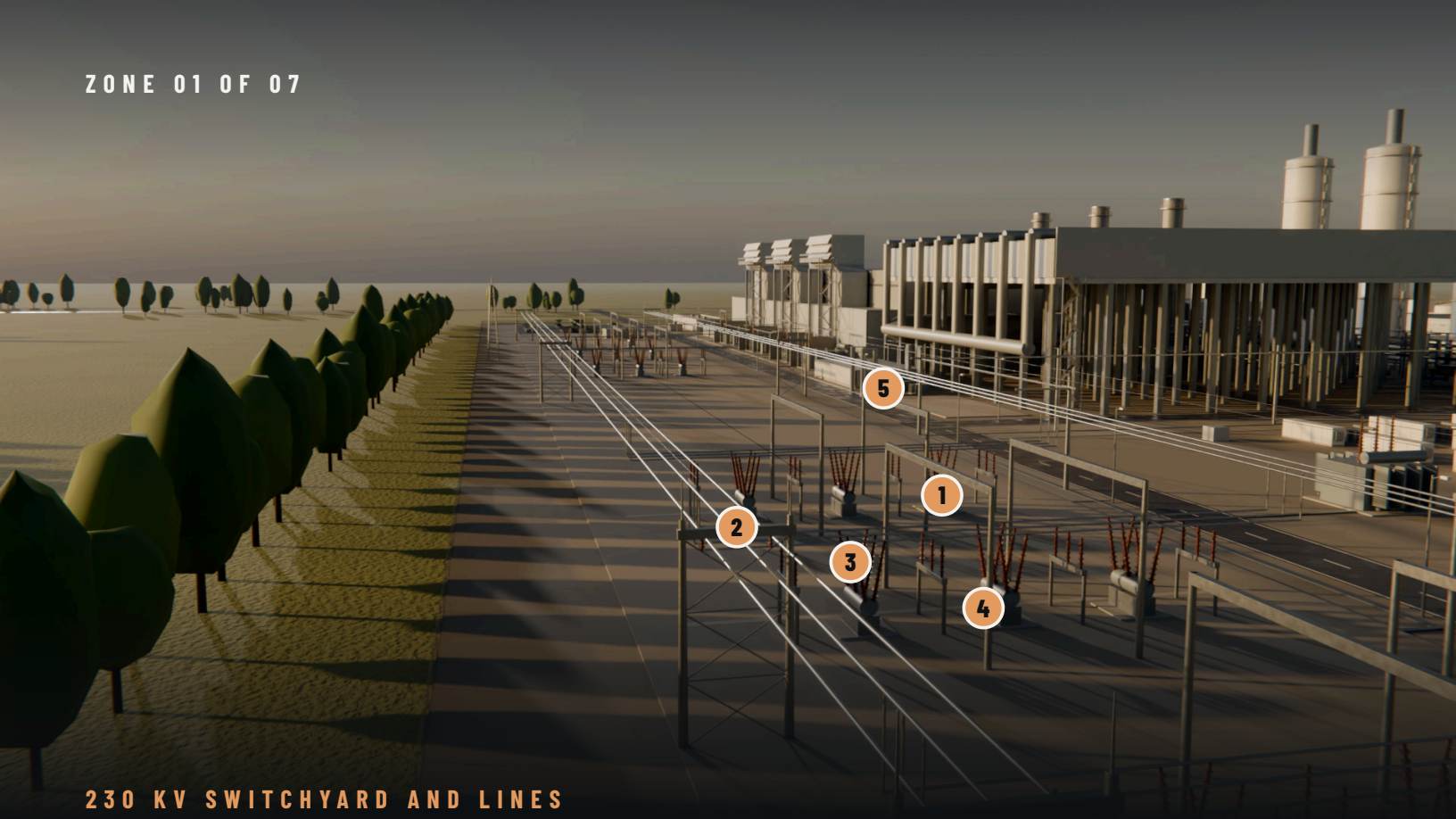
1 PV / RHW-2 2 kV DC conductor
2 MV-105 35 kV aluminium, concentric neutral
3 Type TC-ER 600 V
4 Shielded control / communications cable
5 Bare copper grounding

07 MODULAR AND TEMPORARY POWER

1 Type SHD-GC portable power cable, 15 kV
2 Type W portable power cable, 2 kV
3 MV-105 15 kV in ground trays
4 Type G-GC portable cable
5 Type TC-ER control



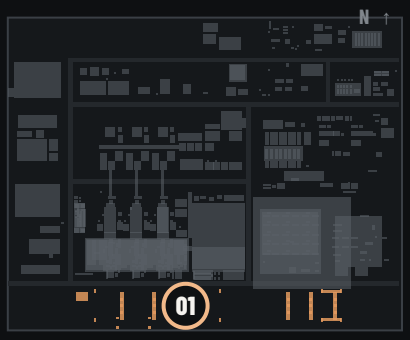
Campus plan drawn from the SK-3X1 3D model: every footprint at true position, zones tinted by model area. Data-centre campus and the marine terminal not shown. Products are typical for each application; confirm against current Southwire specifications.



230 KV SWITCHYARD AND LINES

01 GRID CONNECTION

The plant's output leaves through step-up transformers and a breaker-and-a-half 230 kV switchyard. Where the circuits run underground, HV cable rises to outdoor terminations; overhead conductor carries them to the grid.



WHERE IN THE CAMPUS

1 **HV XLPE TRANSMISSION CABLE, 230 KV**

Underground circuits from the step-up transformers and the CCS and modular-yard ties: Cu conductor, XLPE, metallic moisture barrier, HDPE jacket.

2 **OVERHEAD CONDUCTOR (ACSR / ACSS)**

The 230 kV lines leaving the terminal towers, and the switchyard bus jumpers.

3 **BARE COPPER GROUNDING CONDUCTOR**

The station ground grid, rods and risers under the yard and every structure.

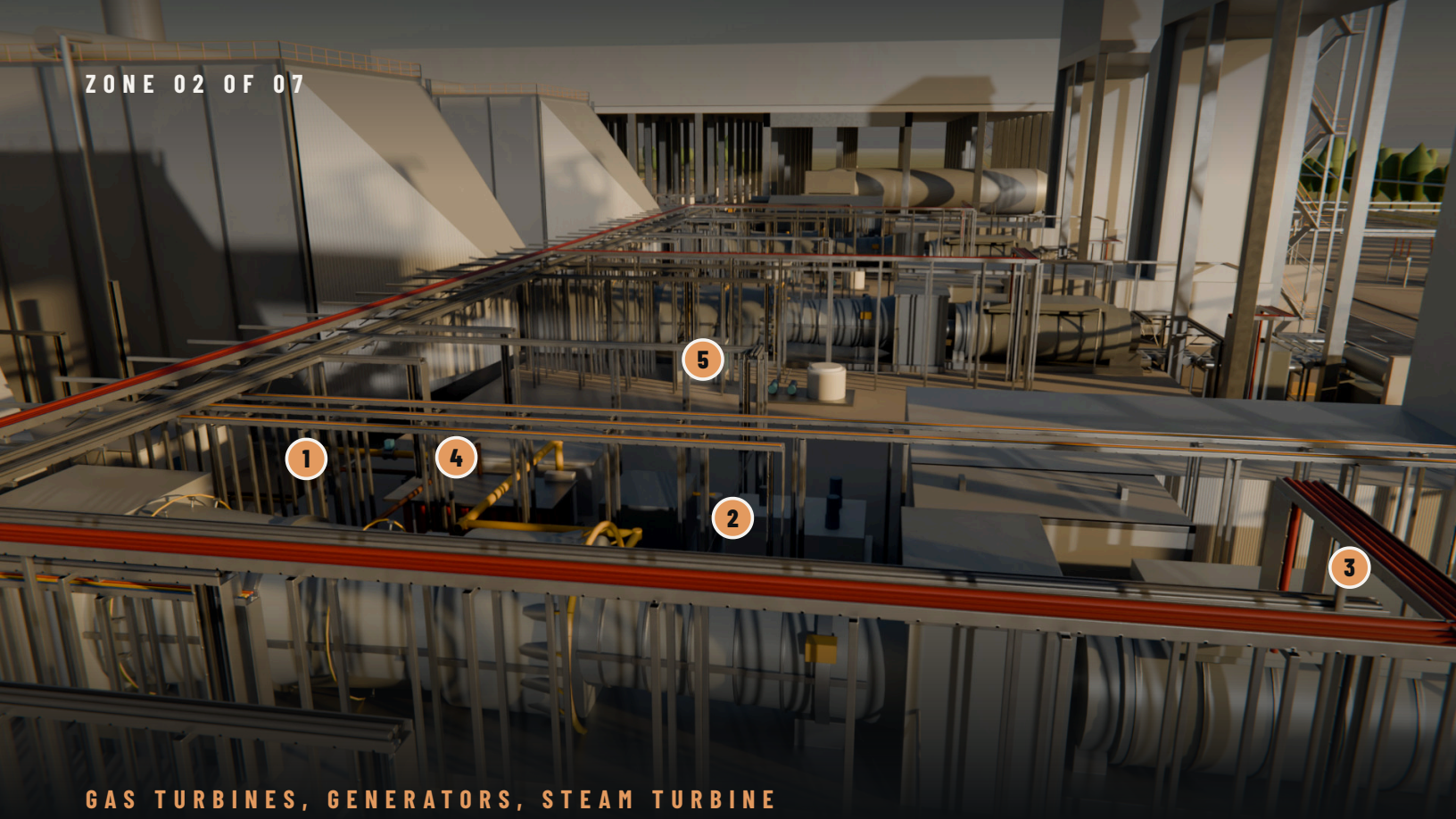
4 **TYPE TC-ER CONTROL CABLE, SHIELDED**

Breaker, CT / VT and relay circuits in the yard trenches to the relay house.

5 **MV-105 SHIELDED POWER CABLE**

15-35 kV station-service and auxiliary feeders at the yard edge.

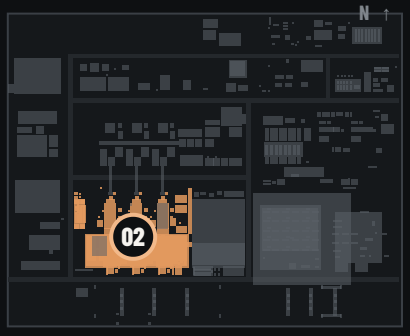
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GAS TURBINES, GENERATORS, STEAM TURBINE

02 POWER ISLAND

Three H-class gas turbines and a steam turbine share one hall. Every auxiliary that keeps them turning, from lube oil to fuel valves to the static starters, is fed and controlled by cable.



WHERE IN THE CAMPUS

1 **ARMOR-X® MC-HL 15 KV, RED PVC**

13.8 kV feeders to auxiliary drives through the Class I, Div. 2 turbine areas, laid in tray with no conduit.

2 **ARMOR-X® MC-HL 600 V, BLACK PVC**

Motor and power circuits on the turbine enclosures, lube-oil and fuel skids.

3 **MV-105 15 KV, EPR 133%**

Switchgear to unit transformers, boiler feed pumps and the static starters.

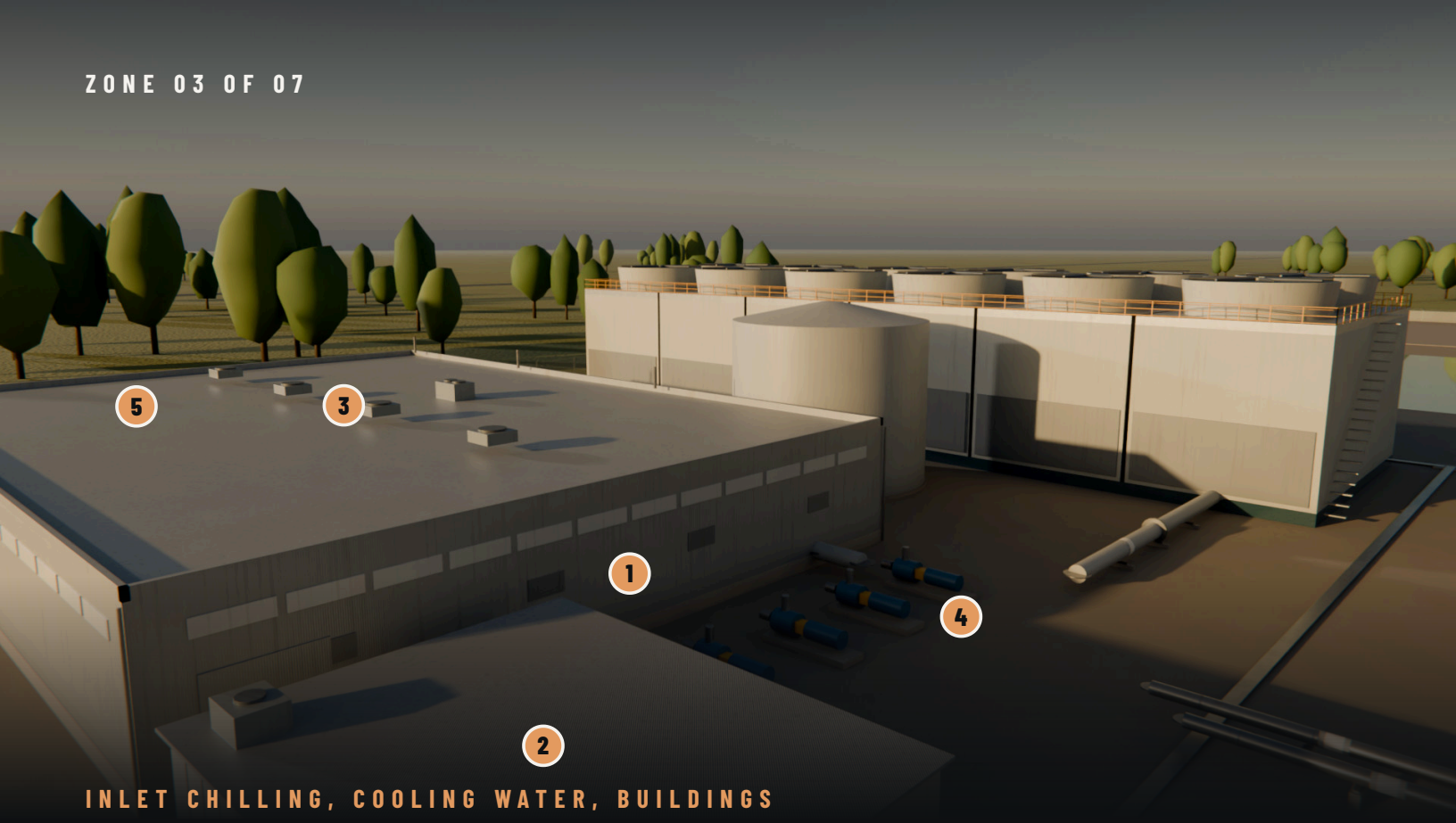
4 **VFD CABLE, SYMMETRICAL GROUNDS, SHIELDED**

Drives to pumps and fans: controls noise and reflected-wave stress.

5 **INSTRUMENTATION CABLE (PLTC / ITC)**

Turbine and HRSG control loops, transmitters and thermocouples.

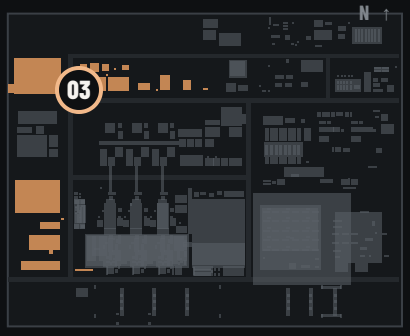
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INLET CHILLING, COOLING WATER, BUILDINGS

03 BALANCE OF PLANT

Around the power island, the balance of plant keeps it running. Here, the inlet-chilling plant: six water-cooled chillers, cooling-water pumps, an e-house and a 12-cell tower, all fed, controlled and protected by cable.



WHERE IN THE CAMPUS

1. **MV-105 5 KV AND 15 KV**
   13.8 and 4.16 kV feeders to the chiller compressors, cooling-water pumps and station transformers.

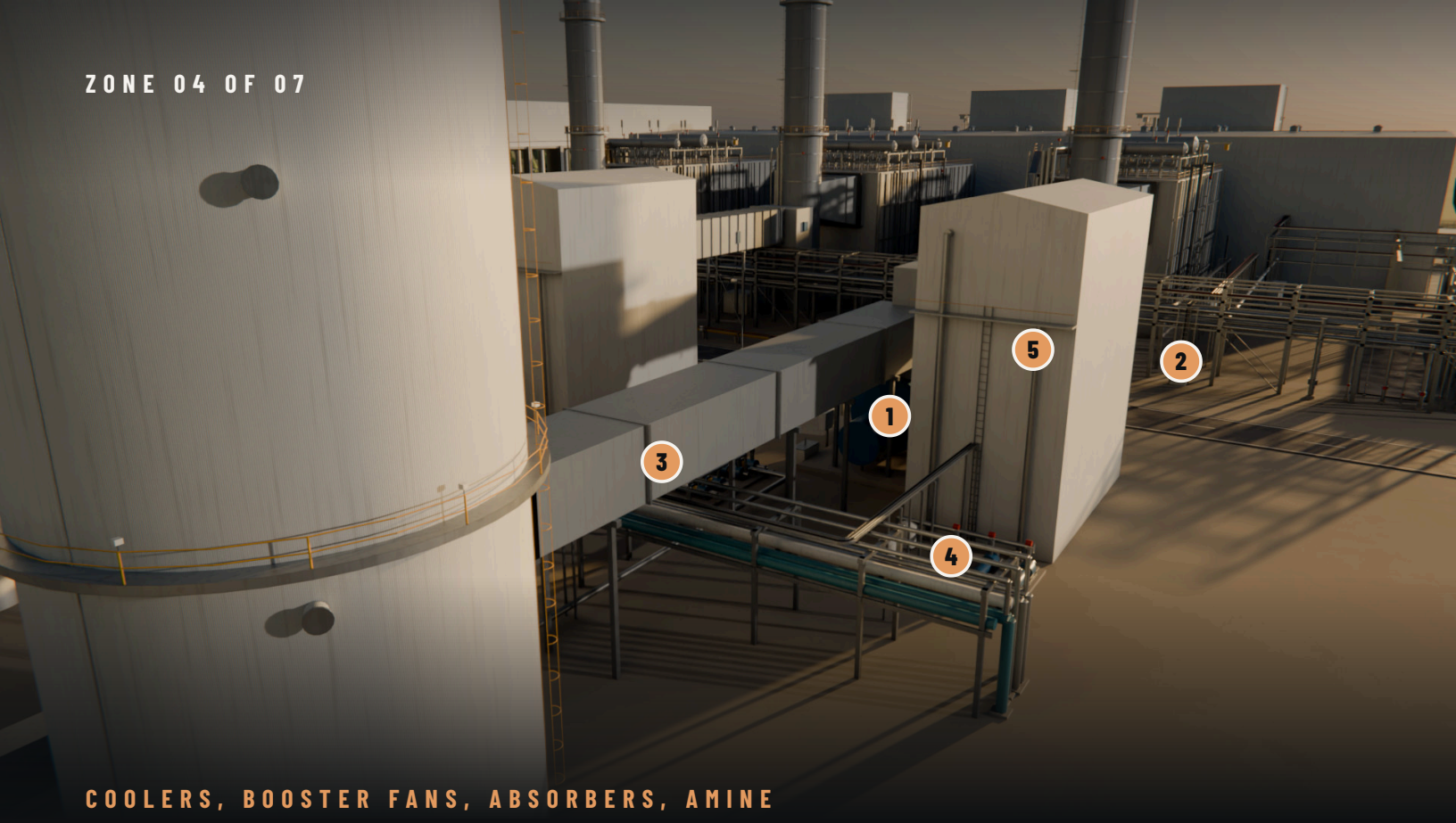
2. **TYPE TC-ER 600 V POWER**
   Motor-control-center and panel feeders out of the e-house and across the racks.

3. **SIMPULL® THHN / XHHW-2 BUILDING WIRE**
   Lighting, receptacles and HVAC in the chiller, water-treatment, control and admin buildings.

4. **TYPE TC-ER CONTROL, MULTICONDUCTOR**
   Pump, valve and level interlocks between the e-house and the process.

5. **FIRE ALARM AND LIFE-SAFETY CABLE**
   Detection and notification circuits in the plant buildings and e-houses.

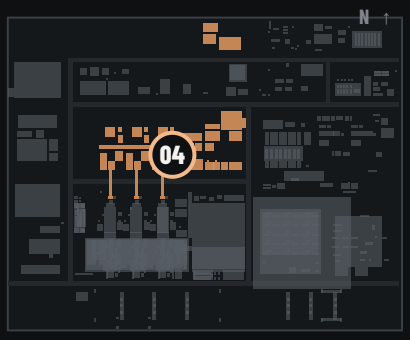
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COOLERS, BOOSTER FANS, ABSORBERS, AMINE

04 CARBON CAPTURE

Post-combustion capture adds a second plant's worth of motors: 16 MW booster fans, quench and solvent pumps, compressors, all fed from the CCS rack.



WHERE IN THE CAMPUS

1 MV-105 15 KV
Feeders to the 16 MW booster fans and the CO2 compressors.

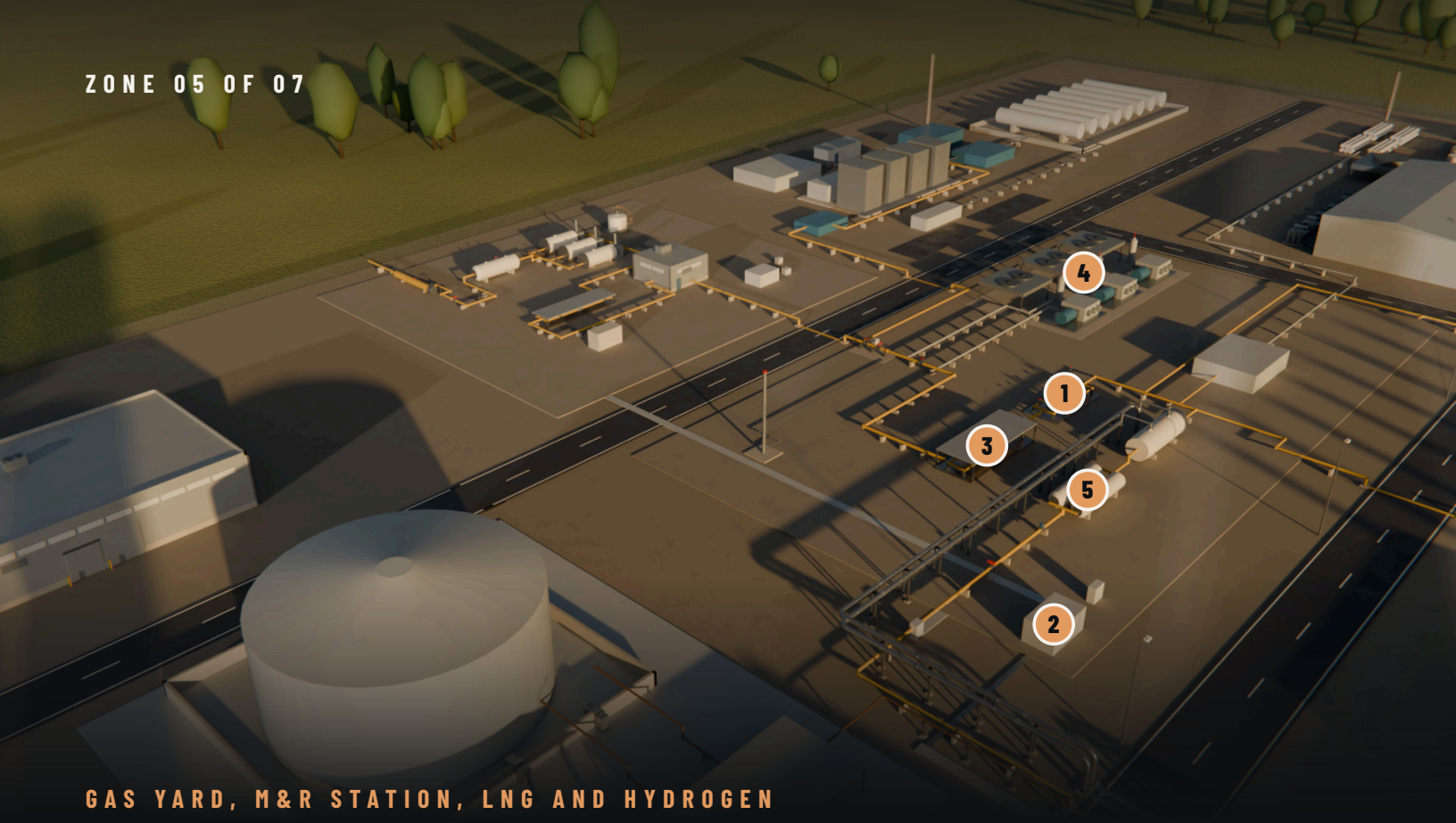
2 ARMOR-X® MC-HL 15 KV
MV circuits along the carbon-capture rack.

3 ARMOR-X® MC-HL 600 V
Pump, blower and actuator power in the amine process area.

4 VFD CABLE
Variable-speed drives on quench and solvent pumps.

5 INSTRUMENTATION CABLE (PLTC / ITC)
Cooler, absorber, stripper and regeneration control loops.

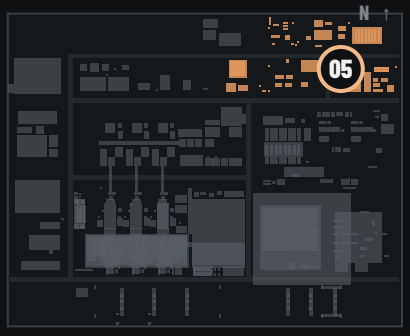
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GAS YARD, M&R STATION, LNG AND HYDROGEN

05 FUEL

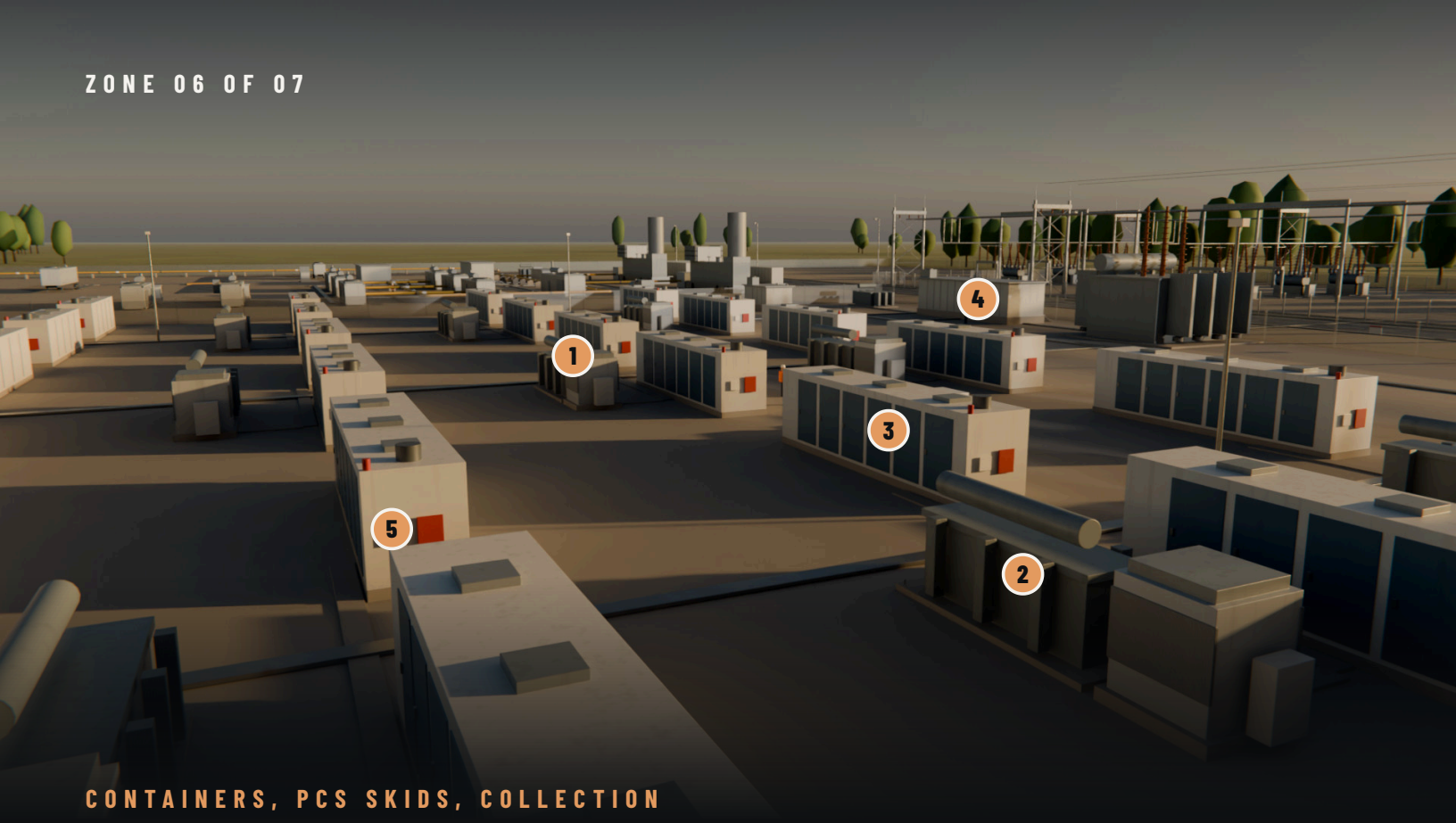
Gas arrives at the metering and regulating station and is filtered, heated and regulated in the plant gas yard: hazardous areas where every circuit is built for them.



WHERE IN THE CAMPUS

1. **ARMOR-X® MC-HL 600 V**
   Power in the Class I areas: regulation and heating skids, vaporizers, electrolyzers.
2. **TYPE TC-ER-HL POWER AND CONTROL**
   Hazardous-location circuits where armor is not required.
3. **ARMORED INSTRUMENTATION CABLE**
   Metering, pressure and gas-detection loops.
4. **MV-105 5-15 KV**
   Fuel-gas compressor and boil-off compressor motors.
5. **BARE COPPER GROUNDING AND BONDING**
   Static and lightning protection at skids, tanks and unloading arms.

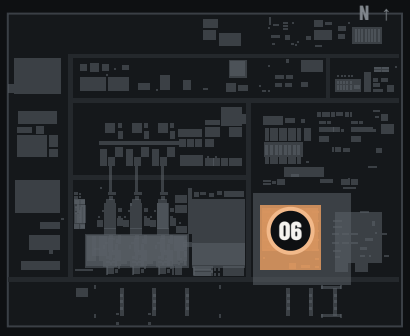
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CONTAINERS, PCS SKIDS, COLLECTION

06 BATTERY STORAGE

The battery yard pairs containers with power-conversion skids. DC conductors run in precast trenches; 35 kV collection loops carry the power to the yard substation.



WHERE IN THE CAMPUS

1. **PV / RHW-2 2 KV DC CONDUCTOR**
   Battery racks to inverters in precast trenches (1,500 V systems).
2. **MV-105 35 KV ALUMINIUM, CONCENTRIC NEUTRAL**
   Collection loops from the PCS / MV skids to the collector e-house.
3. **TYPE TC-ER 600 V**
   Container HVAC, auxiliary power and the energy management system.
4. **SHIELDED CONTROL / COMMUNICATIONS CABLE**
   Battery management, inverter and SCADA links.
5. **BARE COPPER GROUNDING**
   Container and skid grounding to the yard grid.

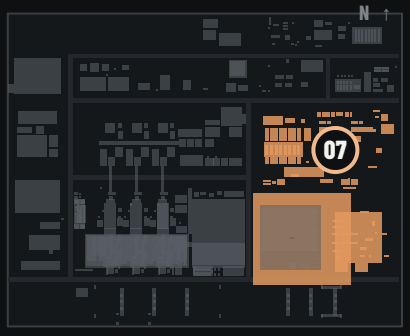
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GENSETS, TRAILER TURBINES, PORTABLE PAD

07 MODULAR AND TEMPORARY POWER

Containerized gensets, trailer turbines and bridge-power fleets come and go. Their cable lies on the pad, crosses lanes under drive-over protectors and plugs in with cam-locks.



WHERE IN THE CAMPUS

1 **TYPE SHD-GC PORTABLE POWER CABLE, 15 KV**
MV feeders from gensets and trailer units, laid on the pad and under drive-over protectors.

2 **TYPE W PORTABLE POWER CABLE, 2 KV**
480 V single conductors with cam-lock connections.

3 **MV-105 15 KV IN GROUND TRAYS**
Bridge-power feeders and TM2500 generator leads to the duct bank.

4 **TYPE G-GC PORTABLE CABLE**
Multi-conductor portable links for smaller gensets and load banks.

5 **TYPE TC-ER CONTROL**
Genset paralleling, protection and fuel-skid control.

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